

BanHalluEval: Human Annotation Guidelines

This document contains the complete annotation guidelines for validating the

BanHalluEval benchmark. It covers five annotation tasks. Read **Section 1 (General Instructions)** first, because it applies to every task. Then read the section for the specific task you have been assigned, and keep it open for reference while you work.

Contents

Task	Section	What you label	Label column(s)	Scale
A	Question Answering (QA)	Whether a Bengali QA answer is hallucinated	is_hallucinated	Yes / No
B	Summarization	Whether a Bengali summary is hallucinated	is_hallucinated	Yes / No
C	Mathematical Reasoning	Whether a Bengali math reasoning chain is hallucinated	is_hallucinated	Yes / No
D	Code-mixed QA	Whether a code-mixed (Banglish) QA answer is hallucinated	is_hallucinated	Yes / No
E	Code-mixed Conversion Quality	Whether a Bengali to code-mixed conversion is faithful and natural	meaning_preserved, naturalness_1to5	Yes / No, and 1 to 5

1. General Instructions

These rules apply to **every** task in this document.

1. **Label every row.** Never leave a row blank. Do not skip a row because it looks hard. If you are unsure, make your best judgment and then **flag** it (see Section 8).
2. **Work independently.** Do not discuss individual items with the other annotators while labelling. Agreement between independent annotators is what makes the benchmark trustworthy.
3. **This is a blind task.** You are **not** told whether an item is supposed to be correct or hallucinated. Judge only what is in front of you.
4. **Judge against the source, not against your outside knowledge.** For every task the decision is made *relative to the provided context, document, or problem*, as spelled out in

each task section.

5. **Use only the allowed labels.** For Yes/No columns write exactly YES or NO. For the 1 to 5 scale write a single whole number 1, 2, 3, 4, or 5.
6. **Flag, don't guess silently.** If a row is genuinely ambiguous, put your best label **and** write a short note in the notes or comment column (or highlight the cell) so a supervisor can review it.
7. **Ask when the guideline does not cover a case.** Contact the supervisor rather than inventing a new rule on the spot, so that all annotators handle the same situation the same way.

What "hallucinated" means

For Tasks A, B, C, and D you are detecting **hallucination**. Across all of them, an output is **hallucinated** when it is **factually incorrect** or **not supported by the source** it is being judged against. Two things follow from this:

- **Added information that is not in the source is a hallucination**, even if it sounds plausible or happens to be true in the real world.
- **Contradicting the source is a hallucination**, whether the output changes a number, a name, a date, a place, or the direction of a claim.

Each task section below refines this general idea with its own rules and examples. (Task E is different: it rates conversion quality rather than detecting hallucination.)

Task A: Question Answering (QA) Hallucination

A.1 Overview

You are given a Bengali **reading-comprehension** item: a source **context** passage, a **question** about that passage, and a **model answer**. Decide whether the model answer is **hallucinated**, meaning whether it contains information that is wrong or that the context does not support.

A.2 What You Will See

Column	Meaning
context	The Bengali source passage. This is the only ground truth.
question	The Bengali question.
answer (model answer)	The answer you must judge.
is_hallucinated	You fill this in: YES or NO.

A.3 Definition

A QA answer is hallucinated when it states something that is **factually incorrect** or **not supported by the context**. The two error patterns you will most often see are:

- **Factualness error.** The answer gives a wrong fact: a wrong name, number, date, place, or relationship compared to what the context says.
- **Comprehension error.** The answer misreads the question or the context: it answers a different question, confuses two entities, or draws a conclusion the passage does not support.

The reference point is always the **context**. If the answer adds specific details (names, dates, figures) that do not appear in and cannot be inferred from the context, treat those added details as unsupported, even if they might sound plausible.

A.4 Labeling Rules

Mark is_hallucinated = Yes if the answer:

- Contradicts a fact stated in the context (e.g. context says 1965, answer says 1975).
- Contains a specific detail (name, number, date, place) that the context neither states nor implies.
- Answers a different question than the one asked, or confuses two people or things in the passage.
- Is partly correct but contains **any** fabricated or wrong piece of information.

Mark is_hallucinated = No if the answer:

- Is fully supported by the context and correctly answers the question.
- Is a paraphrase or reworded version of the correct answer that keeps the same meaning.
- Correctly states that the information is not available, **when the context genuinely does not contain it**.

A.5 Examples

Context: ১৯৬৫ সালের ৩১শে জুলাই ইংল্যান্ডের ইয়েট শহরে জে. কে. রাউলিং জন্মগ্রহণ করেন। তাঁর মায়ের নাম অ্যান। **Question:** জে. কে. রাউলিং-এর মায়ের নাম কী?

- **Answer:** তাঁর মায়ের নাম অ্যান। → **is_hallucinated = No** (*Directly supported by the context.*)
- **Answer:** তাঁর মায়ের নাম মার্গারেট অ্যান। → **is_hallucinated = Yes** (*Adds "মার্গারেট", a name not present in the context, so it is a fabricated detail.*)
- **Answer:** তিনি ১৯৭৫ সালে জন্মগ্রহণ করেন। → **is_hallucinated = Yes** (*Wrong fact, and it also answers the wrong question; the context says 1965, and the question was about the mother's name.*)

Context: পদ্মা সেতুর দৈর্ঘ্য ৬.১৫ কিলোমিটার। **Question:** পদ্মা সেতুর দৈর্ঘ্য কত?

- **Answer:** পদ্মা সেতুর দৈর্ঘ্য প্রায় ৬ কিলোমিটার। → **is_hallucinated = No** (*Correct paraphrase or rounding that preserves the meaning.*)
- **Answer:** পদ্মা সেতুর দৈর্ঘ্য ৯ কিলোমিটার। → **is_hallucinated = Yes** (*Contradicts the number*

in the context.)

A.6 Common Pitfalls

- **"It sounds true, so it must be fine."** Judge against the **context**, not the internet. An added fact that happens to be true in the real world is still a hallucination if the passage does not support it.
 - **Partial correctness.** If any part of the answer is fabricated or wrong, the whole answer is Yes.
 - **Extra harmless wording.** Polite framing, restating the question, or a fuller sentence is fine as long as no new unsupported fact is introduced.
-

Task B: Summarization Hallucination

B.1 Overview

You are given a Bengali source **document** and a **summary** of that document. Decide whether the summary is **hallucinated**, meaning whether it says anything that the document does not support or that contradicts the document.

B.2 What You Will See

Column	Meaning
document	The Bengali source document. This is the only ground truth.
summary	The summary you must judge.
is_hallucinated	You fill this in: Yes or No.

B.3 Definition

A summary is **faithful** when every claim it makes can be traced back to the document. It is **hallucinated** when it introduces content the document does not support. There are two patterns:

- **Intrinsic hallucination.** The summary **contradicts** the document. It misstates a fact that is in the document (wrong number, wrong name, wrong cause and effect, wrong direction of a change).
- **Extrinsic hallucination.** The summary **adds** information that is **not in** the document at all (a fabricated detail, statistic, name, or conclusion).

A good summary may **leave out** details. That is normal and is **not** a hallucination. The problem is only *adding* or *contradicting*, never *omitting*.

B.4 Labeling Rules

Mark **is_hallucinated** = Yes if the summary:

- States a fact that contradicts the document (intrinsic).
- Introduces a detail, number, name, quote, cause, or conclusion that is not in the document (extrinsic).
- Overstates or reverses something in the document (e.g. document says "may reduce", summary says "eliminates"; document says prices *rose*, summary says they *fell*).

Mark `is_hallucinated` = No if the summary:

- Contains only claims that are stated in or directly entailed by the document.
- Is shorter than the document and omits details but adds nothing and contradicts nothing.
- Paraphrases or compresses the document while keeping the facts intact.

B.5 Examples

Document: গত অর্থবছরে দেশের চা রপ্তানি ১২ শতাংশ বৃদ্ধি পেয়েছে। রপ্তানির প্রধান গন্তব্য ছিল যুক্তরাজ্য।

- **Summary:** গত অর্থবছরে চা রপ্তানি বেড়েছে, এবং প্রধান গন্তব্য ছিল যুক্তরাজ্য। → **`is_hallucinated` = No** (*Faithful compression; it omits the exact figure but adds and contradicts nothing.*)
- **Summary:** গত অর্থবছরে চা রপ্তানি ১২ শতাংশ কমেছে। □ **`is_hallucinated` = Yes** (intrinsic) (*Reverses the direction of the change; the document says it rose.*)
- **Summary:** গত অর্থবছরে চা রপ্তানি ১২ শতাংশ বৃদ্ধি পেয়েছে এবং সরকার নতুন কর ছাড় ঘোষণা করেছে। → **`is_hallucinated` = Yes** (extrinsic) (*The tax-break clause appears nowhere in the document, so it is a fabricated addition.*)

Document: নতুন ওষুধটি পরীক্ষায় কিছু রোগীর উপসর্গ কমাতে সাহায্য করতে পারে বলে গবেষকরা জানিয়েছেন।

- **Summary:** গবেষকরা জানিয়েছেন নতুন ওষুধটি কিছু রোগীর উপসর্গ কমাতে সাহায্য করতে পারে। → **`is_hallucinated` = No**
- **Summary:** নতুন ওষুধটি সব রোগীকে সম্পূর্ণ সুস্থ করে তোলে। → **`is_hallucinated` = Yes** (*Overstates "may help some patients" into "cures all patients", which is unsupported and contradictory.*)

B.6 Common Pitfalls

- **Omission is not hallucination.** A summary that drops details is fine. Only mark YES for added or contradicted content.
 - **Watch the modifiers.** "may / some / partly" versus "will / all / completely" changes meaning. Strengthening a hedged claim into a certain one is a hallucination.
 - **Plausible is not supported.** A fabricated statistic or name is still a hallucination even if it sounds reasonable. Check that it is actually in the document.
-

Task C: Mathematical Reasoning Hallucination

C.1 Overview

You are given a Bengali **mathematical word problem** (from the SOMADHAN dataset), a step-by-step **reasoning chain** that solves it, and the **final answer** produced by that chain. Decide whether the reasoning chain is **hallucinated**, meaning whether it contains any incorrect step, fabricated assumption, or wrong conclusion.

C.2 What You Will See

Column	Meaning
question	The Bengali math word problem.
hallucinated_chain (reasoning chain)	The step-by-step solution you must judge.
answer / final answer	The final result the chain arrives at.
is_hallucinated	You fill this in: Yes or No.

Important: Evaluate the **whole chain**, not just the final answer. A chain can reach a number that looks reasonable while containing a wrong step, and a chain with a correct final number can still be hallucinated if it got there through a flawed step. Check every step.

C.3 Definition

A reasoning chain is hallucinated if **any** of the following is true:

Error type	What it looks like
Arithmetic slip	A single calculation is wrong (e.g. ২৫×৮ written as ১৮০ instead of ২০০).
Formula misapplication	The wrong operation is used (divides instead of multiplies, adds instead of subtracts), often with wording that tries to justify it.
Variable confusion	Two quantities are swapped (e.g. uses the manager's rate where the worker's rate belongs).
Invalid deduction	A conclusion is drawn that does not logically follow from the previous step.
Hallucinated intermediate fact	An assumption is introduced that is nowhere in the problem (e.g. "workers get 2 days off" when the problem never says so).
Semantic drift	The chain quietly changes what the problem is asking (e.g. the problem asks for wages plus tax, but the chain answers wages alone).

If none of these occurs and the final answer is correct, the chain is **not** hallucinated.

C.4 Labeling Rules

Mark is_hallucinated = Yes if:

- Any arithmetic step is computed incorrectly.
- Any step uses a wrong operation or a wrong or swapped quantity.
- The chain assumes a fact that is not stated in the problem.
- A step's conclusion does not follow from what came before.
- The chain answers a different question than the one asked (drift).
- The final answer is wrong, **or** it is right but reached through a flawed step.

Mark is_hallucinated = No if:

- Every arithmetic step is correct.
- Every operation and every quantity used matches the problem.
- No fact is used that the problem does not provide.
- Each step follows logically from the previous one.
- The final answer correctly answers exactly what was asked.

C.5 How to Check a Chain (recommended procedure)

1. Read the problem and note **what is being asked** and **what numbers or quantities** are given.
2. Go step by step. For each step, verify: *Is the arithmetic right? Is the right operation used? Are the right quantities used? Does this follow from the last step? Is every fact from the problem?*
3. Confirm the final answer both **matches the chain** and **answers the actual question**.
4. The **first** violation you find is enough to mark YES. You do not have to catalogue them all.

C.6 Examples

Problem: একজন শ্রমিক ২৫ দিন কাজ করেন, প্রতিদিন ৮ ঘণ্টা। মোট কত ঘণ্টা কাজ করলেন?

- **Chain:** $২৫ \text{ দিন} \times ৮ \text{ ঘণ্টা/দিন} = ২০০ \text{ ঘণ্টা}$ । উত্তর: ২০০ ঘণ্টা। $\square$ **is_hallucinated = No** (Correct operation, correct arithmetic, answers what was asked.)
- **Chain:** $২৫ \text{ দিন} \times ৮ \text{ ঘণ্টা/দিন} = ১৮০ \text{ ঘণ্টা}$ । উত্তর: ১৮০ ঘণ্টা। $\rightarrow$ **is_hallucinated = Yes** (Arithmetic slip: $২৫ \times ৮ = ২০০$, not ১৮০.)
- **Chain:** মোট ঘণ্টা বের করতে দিনকে ঘণ্টা দিয়ে ভাগ করি: $২৫ \div ৮ = ৩.১২৫ \text{ ঘণ্টা}$ । $\rightarrow$ **is_hallucinated = Yes** (Formula misapplication: division instead of multiplication.)
- **Chain:** প্রশ্নে বলা আছে শ্রমিক মাসে ২ দিন ছুটি নেন, তাই কার্যকর দিন = $২৫ - ২ = ২৩$; $২৩ \times ৮ = ১৮৪ \text{ ঘণ্টা}$ । $\rightarrow$ **is_hallucinated = Yes** (Hallucinated intermediate fact: the "2 days off")

assumption appears nowhere in the problem.)

C.7 Common Pitfalls

- **Do not judge by the final number alone.** A correct-looking answer can hide a wrong step, and a wrong step makes the chain hallucinated even if the number is right.
 - **Plausible-sounding Bengali is not proof.** Some errors are written to *sound* justified ("তাই ভাগ করি..."). Check the math, not the fluency.
 - **Read what is actually asked.** If the problem wants the total but the chain answers a sub-part, that is semantic drift, so mark Yes.
-

Task D: Code-mixed (Banglish) QA Hallucination

D.1 Overview

This task is the same idea as Task A, but the text is **code-mixed**: Romanised Bengali mixed with English (often called *Banglish*), for example "1965 saler 31 July England-er Yate shohore jonmogrohon koren J.K. Rowling...". You are given a code-mixed **context**, a code-mixed **question**, and a **model answer**. Decide whether the answer is **hallucinated** relative to the context and question.

D.2 What You Will See

Column	Meaning
codemix_context (context)	The code-mixed source passage. This is the only ground truth.
codemix_question (question)	The code-mixed question.
answer (model answer)	The answer you must judge.
is_hallucinated	You fill this in: Yes or No.

D.3 Definition

Exactly as in Task A, an answer is hallucinated when it is **factually incorrect** or **not supported by the context**. The two patterns are:

- **Factualness error.** A wrong fact (wrong name, number, date, place, or relationship compared to the context).
- **Comprehension error.** The answer misreads the question or the context: it answers a different question, or confuses two entities.

Judge everything **relative to the code-mixed context**, regardless of the script or spelling used.

D.4 Reading Code-mixed Text

- **Spelling varies.** The same word may be written many ways (shohor / sohor, koren /

koren). Spelling differences are **not** hallucinations. Judge the **meaning**.

- **Mixed script is normal.** English words, numbers, and names inside Bengali sentences are expected.
- Focus on **what the answer claims**, not on how it is transliterated.

D.5 Labeling Rules

Mark is_hallucinated = Yes if the answer:

- Contradicts a fact in the context (context says one value, answer says another).
- Adds a specific detail (name, number, date, place) not present in or inferable from the context.
- Answers a different question than the one asked, or confuses two people or things.
- Is partly correct but contains **any** fabricated or wrong piece of information.

Mark is_hallucinated = No if the answer:

- Is fully supported by the context and correctly answers the question.
- Is a reworded or re-transliterated version of the correct answer with the same meaning.
- Correctly says the information is unavailable, **when the context genuinely lacks it**.

D.6 Examples

Context: "1965 saler 31 July England-er Yate shohore J.K. Rowling jonmogrohon koren. Tar mayer nam Anni." **Question:** J. K. Rowling-er mayer nam ki?

- **Answer:** Tar mayer nam Anni. → **is_hallucinated = No** (*Directly supported by the context.*)
- **Answer:** J. K. Rowling-er mayer nam chhilo Margaret Anne. → **is_hallucinated = Yes** (*Adds "Margaret Anne", a name not in the context, so it is a fabricated detail.*)
- **Answer:** Tini 1975 sale jonmogrohon koren. → **is_hallucinated = Yes** (*Wrong fact and wrong question; the context says 1965 and the question was about the mother's name.*)

Context: "Padma Bridge-er length 6.15 kilometer." **Question:** Padma Bridge koto lomba?

- **Answer:** Padma Bridge prai 6 km lomba. → **is_hallucinated = No** (*Correct paraphrase or rounding, same meaning.*)
- **Answer:** Padma Bridge 9 km lomba. → **is_hallucinated = Yes** (*Contradicts the number in the context.*)

D.7 Common Pitfalls

- **Do not penalise spelling or transliteration.** sohor versus shohor is not a hallucination. Only the **facts** matter.
 - **Plausible is not supported.** An added fact that sounds right is still a hallucination if the context does not support it.
 - **Partial correctness means Yes.** Any fabricated or wrong piece makes the whole answer Yes.
-

Task E: Code-mixed Conversion Quality

This task is **different from the others**. You are not detecting hallucination. You are rating the **quality of a Bengali to code-mixed conversion** on two independent dimensions.

E.1 Overview

Each item started as an **original Bengali** QA passage, question, or answer, which was then **converted** into Romanised code-mixed Bangla-English text. Judge the conversion on:

1. **meaning_preserved** (Yes / No): does the code-mixed version carry the **same meaning** as the original Bengali?
2. **naturalness_1to5** (1 to 5): how **natural and fluent** does the code-mixed text read, as something a real bilingual Bangladeshi would actually write?

These are two separate judgments. A conversion can preserve the meaning perfectly (Yes) yet read awkwardly (low naturalness), or read smoothly (high naturalness) while dropping or changing meaning (No). Judge each column on its own.

E.2 What You Will See

Column	Meaning
Original Bengali text	The source (context, question, or answer) in Bengali script.
Code-mixed text	The Romanised Banglish conversion you must judge.
meaning_preserved	You fill this in: Yes or No.
naturalness_1to5	You fill this in: a single whole number 1, 2, 3, 4, or 5.

E.3 Dimension 1: meaning_preserved (Yes / No)

The conversion preserves meaning when the code-mixed text says the **same thing** as the original Bengali (the same facts, the same question, the same answer) with nothing important added, dropped, or changed.

Mark meaning_preserved = Yes if:

- Every fact, name, number, and relationship from the original is present and unchanged.
- The question asks for the same thing; the answer gives the same thing.
- Only the script or spelling changed (Bengali to Romanised), not the content.

Mark meaning_preserved = No if:

- Any fact, name, number, or date is dropped, added, or changed.
- The question or answer now means something different.
- A word is mistranslated so the sense changes (e.g. "সী" rendered as "sister").
- Part of the original content is missing.

Spelling and transliteration choices do not affect this column. Only whether the **meaning** is the same matters here.

E.4 Dimension 2: naturalness_1to5 (1 to 5 scale)

Naturalness is how much the code-mixed text reads like genuine, everyday Banglish that a bilingual Bangladeshi would actually write, rather than stiff, machine-like, or word-for-word text. Rate **only the fluency and style**, independently of whether the meaning is correct.

Score	Meaning	What it reads like
5	Completely natural	Reads exactly like real Banglish; you would not guess it was converted.
4	Mostly natural	Fluent with a slight awkwardness or one stiff word choice.
3	Acceptable but mixed	Understandable, but noticeably clunky in places; some unnatural phrasing.
2	Largely unnatural	Word-for-word or machine-like; awkward ordering; hard to read smoothly.
1	Very unnatural	Broken, garbled, or barely readable as Banglish.

Guidance:

- Judge how a **real bilingual speaker** would react, not a strict grammarian.
- Awkward word order, over-literal transliteration, and forced English substitutions **lower** the score.
- Minor spelling variation that people commonly use does **not** lower the score. Natural Banglish is spelled inconsistently.

E.5 Examples

Original Bengali: ১৯৬৫ সালে জে. কে. রাউলিং ইংল্যান্ডে জন্মগ্রহণ করেন।

- **Conversion:** 1965 sale J. K. Rowling England-e jonmogrohon koren. → meaning_preserved = Yes, naturalness_1to5 = 5 (*Same meaning; reads like natural Banglish.*)
- **Conversion:** 1965 shongkhok bochore J. K. Rowling England namok deshe jonmo laav koren. → meaning_preserved = Yes, naturalness_1to5 = 2 (*Meaning is intact, but the phrasing is stiff and over-formal, not how people actually write.*)
- **Conversion:** 1975 sale J. K. Rowling America-te jonmogrohon koren. → meaning_preserved = No, naturalness_1to5 = 5 (*Reads naturally, but the year and the country were both changed, so the meaning is broken.*)
- **Conversion:** J. K. Rowling born hoise ek jaygায়ek somoy. → meaning_preserved = No, naturalness_1to5 = 1 (*Drops the year and place, and the text is garbled, so it fails on both dimensions.*)

E.6 Common Pitfalls

- **Keep the two columns separate.** Do not lower naturalness because the meaning is wrong, and do not mark meaning NO just because the style is awkward.
 - **Don't punish normal Banglish spelling.** Inconsistent spelling is authentic and should not reduce either score.
 - **Small omissions still break meaning.** A dropped number or name is enough for meaning_preserved = No, even if the sentence otherwise reads well.
-

8. Handling Uncertainty and Contact

- **Label every row.** Do not leave any row blank for any task.
- **When genuinely unsure,** give your best label and add a short note (or highlight the cell) so a supervisor can review it. For the naturalness scale, if you are torn between two adjacent scores, pick the lower one and flag it.
- **Work independently,** and contact the supervisor for any case these guidelines do not cover, so that every annotator handles the same situation the same way.

Point of contact

For any confusion, questions, or ambiguous cases, please contact the corresponding person for this annotation through the shared annotation channel provided to you.

Do not resolve unclear cases on your own. Reach out to the corresponding person so that all annotators stay consistent with one another.

Consistent application of these guidelines is what makes the BanHalluEval agreement scores reliable. Thank you for your careful work.